

Himanshu Kaushik

Department of Chemistry,

University of Delhi,

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Languages: Hindi (Native), English (Fluent)

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SUMMARY

Motivated and detail-oriented intern with a Bachelor's and Master's in Chemistry from the University of Delhi, specializing in organic synthesis. Eager to pursue a PhD in Chemistry, with a passion for research and a drive to contribute to groundbreaking discoveries in the field. I thrive in diverse environments, both in the lab and beyond, driven by a curiosity for the world around me.

RESEARCH

Research Intern

(June 2024 – Present)

Advisor: Dr. Snigdha Singh, Drug Discovery and Development Laboratory, Department of Chemistry, University of Delhi, Delhi, India.

Project Title: "Synthesis of novel 8-hydroxyquinoline analogs for biomedical applications"

I am currently engaged in the synthesis of novel derivatives of 8-hydroxyquinoline, which have promising applications in metal sensing. Our research encompasses the evaluation of metal-binding affinities and the fluorescence responses of these derivatives post-synthesis. Our objective is to develop effective fluorescent sensors capable of detecting metal ions across various domains, including medical applications and the mitigation of environmental hazards associated with toxic metals. Additionally, I will contribute to the preparation of a compound library utilizing the Betti reaction and will conduct tests against various cancer cell lines.

Dissertation

(Jan 2024 - May 2024)

Advisor: Prof. A.K Verma, Prof. S. Kukreti, Prof. S.K Awasthi, Prof. R.K. Parashar

Literature review on "Chemical potential of strained ring systems to get driven towards formal Cycloadditions by homolytic opening"

BOOK CHAPTER

H. Kaushik, Nisha, Yashaswi Sharma, Ratneshwar Kumar Ratnesh, Ramesh Chandra, Snigdha Singh. Chapter: Advancement, Challenges and Applications of Nanomaterials for the book entitled "Nanotechnology Applications for Industry 4.0" by CRC Press, Taylor & Francis.

MANUSCRIPT UNDER PREPARATION

Synthesis of novel 8-hydroxyquinoline analog as fluorescent sensors for the detection of metal ions.

CONFERENCES ATTENDED

60th Annual Convention of Chemists, IIT Delhi Campus

20-21st December, 2023

Theme - Net zero goal and sustainability: Role of chemical sciences in green energy, circular economy and prosperity of India.

EDUCATION

Kirori Mal College, University of Delhi

2022 - 2024

Masters of Science, Chemistry

CGPA- 6.9

Kirori Mal College, University of Delhi

2018 - 2021

Bachelor of Science, Chemistry

CGPA-8.2

GR Senior Secondary School, Bhiwani, Haryana

2017

12th Grade; Percentage-94.6 %

Public Senior Secondary School Bal Bhawan, Bhiwani, Haryana

2015

10th Grade; CGPA-9.6

SKILLS

- **Field Skills:** Organic Synthesis, Spectroscopy Techniques (NMR, IR, UV-Vis, Fluorescence), Chromatography (Column, TLC), Laboratory Techniques (Titration, Crystallization, Distillation).
- **Computer Skills:** ChemDraw, Origin, MestReNova, Biorender, Microsoft Office (Excel, Word, PowerPoint)

REFERENCE

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| 1. Prof. Ramesh Chandra
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